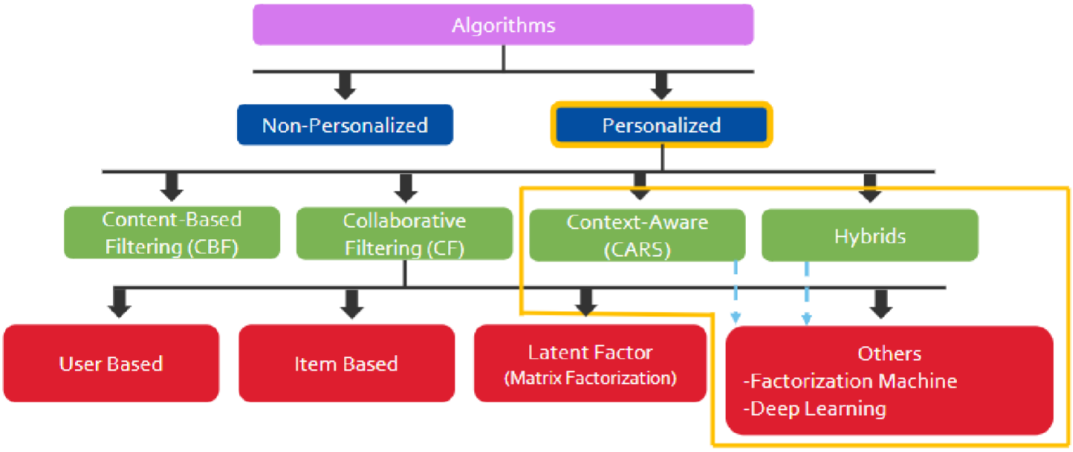
Recommender Systems Algos

Notes on the course Recommender Systems as taught by Paolo Cremonesi and Maurizio Dacrema in Politecnico di Milano during the academic year 2019/2020

Recommender Systems Algos
- Taxonomy
- Non Personalized
  - 1 - Top Popular
  - 4 - Top Rated
- Personalized
  - Global Effects
  - Content Based
    - 1 - IBCBF
    - 2 - IBCBF with TF-IDF
  - Collaborative Filtering
    - 1 - UBCF
    - 2 - IBCF
  - Association Rules
  - Machine Learning Recommenders
    - SLIM, BPR, IALS
      - SLIM
      - BPR
      - Slim BPR
      - IALS (TODO)
    - Matrix Factorization Techniques
      - FunkSVD
      - SVD ++
      - Adapted SVD ++
      - Asymmetric SVD
    - Side Information Recommenders
      - S-SLIM
      - Collaborative Boosted Content Based Filtering
      - Tensor Factorization
    - Factorization Machines
    - Graph Based Recommenders
      - Page Rank

Taxonomy



Non Personalized

1 - Top Popular

Implicit & Explicit

recommend items with the most number of interactions.
s stands for score.

$$s_i = \sum_{u \in U} r_{ui}$$

sort $\bar{s}$ in descending order and take the first k items.
Most Popular items are those who have most ratings whatever is their value. It is good to remove items the user have already seen.

4 - Top Rated

$$\hat{r}_i = \frac{\sum_{u \in U} r_{ui}}{N_i + C}$$

N_i is the number of users that have given a score to item i.

C is the shrinkage.

Personalized

Global Effects

- compute the average rating of all items by all users

  $$r_{avg} = \frac{\sum_{u,i} r_{ui}}{N^+}$$

  where N^+ is the number of interactions.
- subtract if from each rating in the URM

  $$r'_{ui} = r_{ui} - r_{avg} \qquad \forall (u,i) \in \text{Interactions}$$
- compute for each item the item bias

  $$b_i = \frac{\sum_u r'_{ui}}{N_i + C} \qquad \forall i \in I$$

  where N_i is the number of users that interacted with item i
- remove it from each rating in the URM

$$r''_{ui} = r'_{ui} - b_u \qquad \forall (u,i) \in \text{Interactions}$$

- compute for each user the user bias

$$b_u = \frac{\sum_i r''_{ui}}{N_u + C} \qquad \forall u \in U$$

  where N_u is the number of items rated by u.

- remove it from each rating the URM

$$r'''_{ui} = r''_{ui} - b_i \qquad \forall (u,i) \in \text{Interactions}$$

$\hat{r}_{ui} = r_{avg} + b_i + b_u$

Content Based

- Content Based Filtering:
  use the ICM
- Knowledge Based Filtering:
  when the user only buys occasionally so we can't create user profiles (e.g. a consumer electronics shop). We use additional, often manually provided, information about the users and the available items. content not only of items but even of users.

1 - IBCBF

Item Based Content Based Filtering.

Cosine Similarity

$$s_{ij} = \frac{\sum_a i_a j_a}{\sqrt{\sum_a i_a^2 \sum_a j_a^2} + C}$$

where x_{if} tells you if feature f is present in item i.

$$\tilde{r}_{ui} = \frac{\sum_{j \in KNN} r_{uj} s_{ij}}{\sum_{j \in KNN} s_{ij}}$$

2 - IBCBF with TF-IDF

it is useful to use TF-IDF as well:

TF

$$TF = \frac{N_{ai}}{N_i}$$

N_{ai} is the number of times attribute a appears in item i (usually 1).
N_i is the number of attributes of item i.

$$IDF = \log \frac{N_I}{N_a}$$

N_I is the number of items.

N_a is the number of items in which attribute a appears.

$$\text{TF-IDF}(a,i) = TF(a,i) \cdot IDF(a)$$

In the TF-IDF model, the document is represented not as a vector of boolean values for each keyword but a s a vector of the computed TF-IDF weights.

Collaborative Filtering

1 - UBCF

User Based Collaborative Filtering

Implicit

Cosine Similarity

$$s_{uv} = \frac{\sum_i r_{ui} r_{vi}}{\sqrt{\sum_i r_{ui}^2 \sum_i r_{vi}^2} + C}$$

$$\tilde{r}_{ui} = \frac{\sum_{v \in KNN} r_{vi} s_{uv}}{\sum_{v \in KNN} s_{uv}}$$

Explicit

Pearson Similarity

$$s_{uv} = \frac{\sum_i (r_{ui} - r_u)(r_{vi} - r_v)}{\sqrt{\sum_i (r_{ui} - r_u)^2 \sum_i (r_{vi} - r_v)^2} + C}$$

$$\tilde{r}_{ui} = \frac{\sum_{v \in KNN} (r_{vi} - r_v) s_{uv}}{\sum_{v \in KNN} s_{uv}} + r_u$$

2 - IBCF

Implicit

Cosine Similarity

$$s_{ij} = \frac{\sum_u r_{ui} r_{uj}}{\sqrt{\sum_u r_{ui}^2 \sum_u r_{uj}^2} + C}$$

$$\tilde{r}_{ui} = \frac{\sum_{j \in KNN} r_{uj} s_{ij}}{\sum_{j \in KNN} s_{ij}}$$

Explicit Ratings

Adjusted Cosine Similarity

$$s_{ij} = \frac{\sum_u (r_{ui} - r_u)(r_{uj} - r_u)}{\sqrt{\sum_u (r_{ui} - r_u)^2 \sum_u (r_{uj} - r_u)^2} + C}$$

$$\tilde{r}_{ui} = \frac{\sum_{j \in KNN} r_{uj} s_{ij}}{\sum_{j \in KNN} s_{ij}}$$

probably we can avoid to put the bias into the prediction formula because they rule out each other, being considered always r_u.

Association Rules

Support

$$\text{Support}(i, j) = \frac{\text{SupportCount}(i, j)}{N_U}$$

Confidence

$$\text{conf}(i,j) = \frac{\text{SupportCount}(i,j)}{\text{SupportCount}(i)}$$

Similarity as Confidence! - not symmetric

$$P(i|j) = \frac{\# < i,j >}{\# < j > +C} = s_{ij} = \text{conf}(j,i)$$

$$score_{i \in I} = \sum_{\text{r} \in \text{rules recommending } i} \text{support}_r \cdot \text{conf}_r$$

it's a weighted sum of supports, the weights are the confidences.

Generally speaking:

- you compute the support for any pair of items (i,j) and if it is greater or equal to $minsup$, you consider it as a frequent itemset.
- for each frequent itemset (i,j), you consider all the association rules associated to such frequent itemset (in this case $i \to j$, $j \to i$) and for each association rule, if its confidence is greater than $minconf$ it is a valuable association rule.

Machine Learning Recommenders

SLIM, BPR, IALS

SLIM

Sparse Linear Methods.
It is a family of algorithms whose goal is to minimize Residual Sum of Squares.
we can see a recommendation task as a machine learning task in which we need to approximate the weights of a similarity matrix.
the formula that SLIM wants to optimize is

$$L = |R - RS|_2 + \alpha|S|_1 + \beta|S|_F$$

The one above is probably just an example of possible optimization. SLIM refers t the fact that I want to minimize the residual, and add some penalization for overfitting.

Elastic Net

$$L = |R - RS|_2 + \alpha|S|_1 + \beta|S|_2$$

Closed Form Solution

It is possible if only ridge is present as a regularization term.

$$S_j = (R^T R + \beta I)^{-1} R^T \bar{r}_j$$

Ra**t**to **ra**b**b**-m**Ino** **ra**t**t**o **ro**j**o**.

Early Stopping

- Speak about early stopping.
- divide dataset in training, validation, and test set.
- Use MSE on training set, but do early stopping on MAP.

BPR

Bayesian Pair-wise Ranking.

BPR is able to maximize AUC.

AUC plots fallout (x) with respect to recall (y) .

$$\text{Fallout} = \frac{\text{num of non relevant recommended items}}{\text{num of non relevant items}}$$

$$\text{Recall} = \frac{\text{num of relevant items suggested}}{\text{num of relevant items}}$$

BPR works both with implicit and explicit ratings.

Problem: popularity bias

Explicit

- positive rating: from 4 to 5
- negative rating: from 0 to 2
- Consider the user bias! no formula provided but you can figure it out.

Implicit

- positive rating: 1
- negative rating: 0

pick random triplets of user, positive, and negative ratings and optimize through stochastic gradient descent

$$\tilde{R} = RS$$

$$\tilde{x}_{uij} = \tilde{r}^{+}_{ui} - \tilde{r}^{-}_{uj}$$

$$\max \sum_{uij} \log \frac{1}{1 + e^{-x_{uij}}} - ||S||_2$$

Slim BPR

simply, S is computed through Slim. same as above.

$$\tilde{r}_{ui} = \bar{r}_u \bar{S}_i$$

$$\tilde{r}_{uj} = \bar{r}_u \bar{S}_j$$

$$\tilde{x}_{uij} = \tilde{r}_u - \tilde{r}_{uj}$$

IALS (TODO)

Matrix Factorization Techniques

FunkSVD

Matrix Factorization with MSE using ALS and iterating.

we learn $\tilde{R}$ as a product between two matrices A and B.

$A : U \times F$

$B : I \times F$

F is a hyperparameter and represents the number of latent factors.

$$L = ||R - A \cdot B||_2 + \alpha ||A||_? + \beta ||B||_?$$

Check this out as well, could consider Frobenius instead if you want.

$$MSE = \sum_{(u,i) \in N^+} (r_{ui} - \tilde{r}_{ui})^2$$

matrix factorization advantage: there are way less parameters.

use ALS to update weights.

- $\text{num of latent factors } = 1 \rightarrow \text{Top Popular}$
- $\text{num of latent factors} = \infty \rightarrow \text{Overfitting}$

SVD ++

Version of FunkSVD that works better for explicit ratings.

It considers global effects.

$$\tilde{r}_{ui} = \mu + b_u + b_i + A_u \cdot B_i$$

same loss function of FunkSVD, simply put $\tilde{r}_{ui}$ as I've described it.

Adapted SVD ++

problem with SVD++: it is memory based.

let's make it model based.

$$\tilde{r}_{ui} = \mu + b_u + b_i + \Big(R_u \cdot B \Big) \cdot B_i^T$$

$$\tilde{r}_{ui} = \mu + b_u + b_i + \sum_f \Big(\sum_j r_{uj} b_{jf} \Big) b_{if}$$

the product of this matrices with its transpose $(B \cdot B^T)$ returns as a matrix with the same dimensions of a similarity matrix.
There is only one difference between the original SLIM and this kind of SLIM. In the first, matrix S was a generic matrix, so you were learning all possible values. Here, matrix S has been decomposed into the product of two identical rectangular matrices.
The two algorithms are the same if I constrain SLIM to learn a product between two identical rectangular matrices instead of simply S.

Asymmetric SVD

another way to solve the cold start problem, other than Adapted SVD++.

$$\tilde{r}_{ui} = \mu + b_u + b_i + \sum_f \Big(\sum_j r_{uj} Q_{fj} \Big) Y_{fi}$$

it is called asymmetric because $Q^T Y$ is not symmetric anymore.

Asymmetric SVD is model-based and it has a good quality.
It is considered one of the best recommendations techniques at least for dataset with explicit ratings.

Side Information Recommenders

S-SLIM

the basic loss function of S-SLIM is the following

$$\min_S \alpha ||R - RS||_2 + (1 - \alpha) ||F - FS||$$

where $F = ICM$.

the variant with a penalization term is the following:

$$\min_S \left\{ \alpha ||R - RS||_2 + \beta ||F - FS||_2 + \gamma ||S|| \right\}$$

the norm of the regularization term can probably be any kind of norm we want.

Variant of SSLIM

I think the following is just Collaborative Boosting not explained properly.

$$\min_{S_{CF}, S_{CB}} \left\{ \alpha ||R - RS_{CF}||_2 + \beta ||F - FS_{CB}||_2 + \gamma ||S_{CF} - S_{CB}|| \right.$$

the norm of the regularization term can probably be any kind of norm we want.

Collaborative Boosted Content Based Filtering

it is an alternative version of SSLIM.

1. solve a SLIM problem
$$\min_{S^{CF}} ||R - RS^{CF}||_2 + \lambda ||S^{CF}||$$
2. we impose the following:
$S_{ij}^{CB} \neq \sum_f a_{if} a_{jf}$ but $S_{ij}^{CB} = \sum_f a_{if} a_{jf} w_f$
3. we solve
$$\min_{\bar{w}} ||S^{CF} - S^{CB}||$$
4. we discard S^{CB} and we keep the weights vector $\bar{w}$.

we are assuming that S^{CF} is very good.

I think that the fifth step (not specified by the professor) is to solve S-SLIM taking into account an new weighted item content matrix (with the weights learnt).

Tensor Factorization

it is an algorithm belonging to the context-aware recommender systems family. It can be seen as matrix factorization in 3 dimensions.

Now our URM is 3 dimensional (it can have even more dimensions but we'll use this dimensionality since it is the wisest to use).

R can be factorized as $R = A \cdot B \cdot D^T$.

$[A] = U \times F$. It represents how much user u likes latent feature f.

$[B] = F \times I$. It represents how much latent feature f is important to describe item i.

$[D] = F \times C$. It represents how much feature f is relevant in context c.

$$\tilde{R} = \tilde{A} \cdot \tilde{B} \cdot \tilde{D}$$
$$\tilde{r}_{uic} = \sum_f a_{uf} b_{if} d_{cf}$$

we can learn the predicted values for the 3 matrices by minimizing $||R - \tilde{R}||_2$ or $||R - \tilde{R}||_F$.

Or we can use BPR to have the best accuracy as possible.

Add bias for user and items if the original R matrix contains explicit ratings.

Sparsity problem, solution: penalization term and reduce dimensionality as long as you can.

Factorization Machines

we turn the recommending problem into a regression problem.

columns corresponding to users.

columns corresponding to items.

columns corresponding to content.

$$\tilde{r}_k = \omega_0 + \bar{x}_k^T \bar{\omega} + \bar{x}_k^T W \bar{x}_k$$

$\bar{\omega}$ is a vector of weights of dimension $U + I$.

W is a matrix of weights of dimensionality $(U + I) \times (U + I)$. It models how much user u likes item j.

W is very big though, so read the following:

The way it is written in the paper it is the following, stick with it if you want to be sure:

$$\hat{y}(\bar{x}) := w_0 + \sum_{i}^{n} w_i x_i + \sum_{i=1}^{n} \sum_{j=i+1}^{n} < \bar{v}_i, \bar{v}_j > x_i x_j$$

$$n = \text{num cols}, \;\; w_0 \in \mathbb{R}, \;\; \bar{w} \in \mathbb{R}^n, \;\; V \in \mathbb{R}^{n \times k}$$

$< \cdot, \cdot >$ is the dot product of two vectors of size k:

$$< \bar{v}_i, \bar{v}_j >:= \sum_{f=1}^{k} v_{if} v_{jf}$$

a row $\bar{v}_i$ within V describes the i-th variable with k factors. $k \in \mathbb{N}_0^+$ is a hyperparameter that defines the dimensionality of the factorization.

- w_0 is the global bias
- w_i models the strength of the i-th variable
- $w_{i,j} =< \bar{v}_i, \bar{v}_j >$ models the interaction between the i-th and j-th variable. Instead of using an own model parameter $w_{i,j} \in \mathbb{R}$ for each interaction, the FM models the interaction by factorizing it.

FM can be applied to a variety of prediction tasks, among them there are:

- Regression (use MSE or similar loss)
- Binary classification
- Ranking

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Graph Based Recommenders

- we model our data as a graph.
- nodes represent users and items.
- an arc goes from a user node to an item node if such user consumed such item, and viceversa.
- random walk for n steps, where n is odd. usually $n = 3$.
- the graph can be formulated as a matrix A such that $|A| = (U + I) \times (U + i)$.

$$A_{i,j} = \begin{cases} 1 \text{ if } i \text{ consumed } j \text{ or if } i \text{ was consumed by } j \\ 0 \text{ otherwise} \end{cases}$$

- we create a probability matrix P of the same dimension of M that tells us what is the probability of going to node j from node i

$$P_{ij} = \frac{A_{ij}}{\sum_k^{U+I} A_{ik}}$$

- the probability of getting to node j after $n = 3$ steps from node i is

$$P_{ij} = \sum_a \sum_b P_{ia} P_{ab} P_{bj}$$

  [TODO] write it better.

Page Rank

[TODO]